



DELVAC

FREEZE DRYING SYSTEMS



DELVAC
PUMPS PVT. LTD.



Delvac offers a complete range of Freeze Dryers / Lyophilizers and accessories for quick and efficient sample preparation. They meet the stringent requirements for cryodesiccation applications in Govt & Private Research Centres, Educational institutions, Biotechnology, Pharmaceuticals, Food, Environment, and Power,... With 30 years of service excellence, we have supplied over 400 units of freeze dryers throughout India to different sectors which are used for a wide range of applications.

DELVAC LAB - SCALE FREEZE DRYERS

Based on the capacity and condenser temperature, DELVAC's Lab-scale Freeze Dryer base units come in seven variants, *MINI - LYODEL (-50C / -80C)*, *LYODEL (-50C / -80C / -100C)*, and *MAXI - LYODEL (-50C / -80C)*. The common base unit gives you the advantage to lyophilise products with a wide range of accessories like flasks, vials, ampoules and trays with appropriate accessories.



MINI - LYODEL - 1.5kg

- Compact ,Bench-top model
- Multipurpose & Economical
- For Less product volumes
- Trolley - optional



LYODEL - 5kg

- Bench-top model
- Multipurpose
- Medium product volumes
- Trolley - optional



ULTRA - LYODEL

- Ultra low freezing temp
- Organic Solvent Drying
Fully automatic -7"Touch
- Programable Settings
- Built-in Condenser Coil
- Auto Defrost



MAXI - LYODEL - 18kg

- Lab & small scale production
- High product volumes
- Long-running time
- Floor Model

TECHNICAL SPECIFICATION OF LAB SCALE FREEZE DRYERS

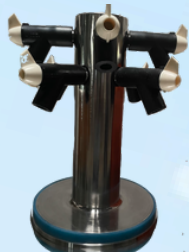
TECHNICAL DATA	MINI - LYODEL		LYODEL		ULRA - LYODEL		MAXI - LYODEL	
PRODUCT CODE/MODEL	LYO1550	LYO1580	LYO0555	LYO0585	ULYO 6	ULYO 12	LYO1855	LYO1885
ULT LOW TEMP +/-5C@20C amb	-50	-80	-55	-80	-105		-50	-85
TYPE	BENCHTOP			FLOOR	BENCHTOP		FLOOR	
CONDENSOR AREA (sq.m)	0.07		0.15		-	-	0.40	
CONDENSER VOLUME (L)	2.3		7.2		15	21	44.5	
ICE CAPACITY 24hr (kg)	1.2		2.3		6	8	10.5	
MAX ICE CAPACITY (kg)	1.5		5.0		8	12	18.0	
CONDENSER MOC	SS 304							
CONDENSER POSITION	VERTICAL		VERTICAL				HORIZONTAL	
COMPRESSOR (hp)	1/4	2 X 1/4	1/3	2 X 1/3			3/4	2 X 3/4
REFRIGERANT	CFC & HCFC FREE							
VAC & TEMP INDICATION	BASIC DIGITAL PIRANI GAUGE / HMI TOUCH SCREEN DISPLAY							
VACUUM RANGE (mbar)	1000 - 0.100							
VACUUM PUMP (LPM)	100		200				250	
DIMENSION - W x D x H (cm)	54 x 36 x 62	85 x 90 x 70	56 x 36 x 82	75 x 90 x 85	780 x 30 x 550	900 x 300 x 720	85 x 95 x 85	95 x 95 x 95
WEIGHT (kg)	50	95	85	140	85	130	225	275
FOOT SPACE (cm)	55 x 55	65 x 85	55 x 75	75 x 85			82 x 95	100 x 100
INPUT POWER SUPPLY	220V, 50HZ, 1 PHASE							

DELVAC LAB SCALE FREEZE DRYER ACCESSORIES

ACCESSORIES FOR LYOPHILISATION IN FLASKS

COLOUMN MANIFOLDS

Column manifolds are suitable for lyophilising small amounts of liquid in flasks. It is supplied with quick seal valves, rubber adaptors, cone connectors, round bottom flasks(max 500ml) and a gasket. MOC - SS 304



PRODUCT CODE	ITEM	PORTS
FM 0080	8 - port SS Column Manifold	8
FM 0024	24 - port SS Column Manifold	24



DRUM MANIFOLDS

Drum manifolds are preferred where simultaneous lyophilisation of liquids in flasks and trays is required. Trays and beakers for bulk drying can be placed inside the drum. It is supplied with quick seal valves, rubber adaptors, cone connectors, round bottom flasks(max 500ml) and gaskets. The material of construction is SS 304/Acrylic.

8 - Port Acrylic Drum - Economical, Transparent, Lightweight
SS Drum Manifold. - High chemical resistance, longer lifespan

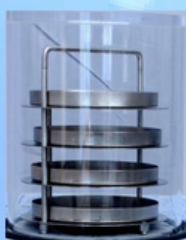
PRODUCT CODE	ITEM	PORTS
FM 0081	8 - Port Acrylic Drum Manifold	8
FM 0008	8 - Port SS Drum Manifold	8
FM 0010	10 - Port SS Drum Manifold	10
FM 0012	12 - Port SS Drum Manifold	12
FM 0016	16 - Port SS Drum Manifold	16



ACCESSORIES FOR LYOPHILISATION IN TRAYS

BULK RACK ASSEMBLY

Bulk rack assembly is an appropriate accessory for freeze-drying of liquids, semi-solids and solids in trays. It comes with 2/3/5 SS racks fitting SS trays, an acrylic jar for closure and a gasket. MOC - SS 304.



PRODUCT CODE	ITEM	TRAYS
BA 02200	2 trays Bulk rack assembly	2
BA 03200	3 trays Bulk rack assembly	3
BA 05200	5 trays Bulk rack assembly	5



ACCESSORIES FOR LYOPHILISATION IN VIALS

VIAL STOPPERING MECHANISM

Vial stoppering is an ideal accessory for samples that require freeze drying in vials followed by sealing under a vacuum. This accessory can be also used for bulk drying. The item comes with 1/2/3 movable aluminium shelves, two semicircular steel trays per shelf and a vacuum chamber. The stoppering mechanism is manual.

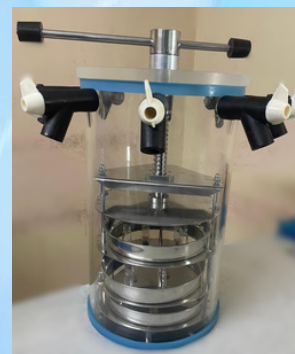
PRODUCT CODE	ITEM	Vial - 2ml	Vial - 5ml	Vial - 10ml
VS 0012	Vial stoppering -1 shelf, 2 trays	80 no	40 no	20 no
VS 0024	Vial stoppering -2 shelf, 4 trays	160 no	80 no	40 no
VS 0036	Vial stoppering -3 shelf, 6 trays	240 no	120 no	60 no

INTEGRATED ACRYLIC DRUM MANIFOLD

The accessory is designed as an integration of a flask manifold and bulk tray with a Vial stoppering mechanism focused on multiple applications in one manifold for easy use for customers. The system consists of standard 8 ports for the flask, with 3 - 5 shelves for vials (2ml, 5ml 10ml) and bulk samples.

Tray dimension Dia 200 ; Height 30

PRODUCT CODE	PORTS	SHELVES	SPACE BET SHELVES	Vial - 2ml	Vial - 5ml	Vial - 10 ml
FVB 083	8	3	70 mm	690 no	360 no	195 no
FVB 084	8	4	50 mm	920 no	480 no	260 no

**HEATER CONTROL UNIT FOR BULK RACK & VIAL STOPPERING**

A heater Control Unit (HCU) is a free-standing power supply unit designed to deliver thermostatically controlled low-voltage power supply to the heater pads attached under each wired tray in a bulk rack assembly/vial stoppering mechanism. The heat energy produced by the heater pads enhances the drying rate. The complete set comes with the heater control box, bulk rack assembly / vial stoppering mechanism.

PRODUCT CODE	ITEM	POWER
BHCU 03300	HCU with 3 trays Bulk rack assembly	22 W/shelf
BHCU 04300	HCU with 4 trays Bulk rack assembly	
BHCU 05300	HCU with 5 trays Bulk rack assembly	
VHCU 0036	HCU with 3 shelf vial stoppering mechanism	
VHCU 0048	HCU with 4 shelf vial stoppering mechanism	

**ACCESSORIES FOR LYOPHILISATION IN AMPOULES****SPIN FREEZER - FREEZING & PRIMARY DRYING**

Spin Freezer was devised to aim for both freezing and primary drying of products in ampoules. The centrifugal action with freezing will expedite the drying process in ampoules by freezing samples to a thin layer on one side of the ampoules. The freezing of ampoules is then followed by primary drying and secondary drying. Sublimation of ice takes place under a vacuum during the primary drying stage and is followed by the secondary drying stage in the secondary drying manifolds.

The spin freeze set includes a control box with a base plate, an acrylic chamber with a gasket, and 5 sets of interchangeable aluminium carriers, each designed to handle different-sized ampoules.



Ampoules (ml)	0.5	1	2	5	10
Capacity (no)	96	80	36	24	12

PRODUCT CODE	ITEM	RPM	POWER
ADSP 096	Spin Freezer with 5 interchangeable carries	500	220V, 50Hz, 1ph

SECONDARY DRYING MANIFOLDS

The Secondary drying manifolds are used for the secondary drying stage after the primary drying of ampoules. In the secondary drying, the lyophilization cycle of ampoules gets completed and the ampoules are sealed under a vacuum with the help of a sealing torch.

The Secondary drying manifold set includes

ADSM 024/048 - 24/48 port manifold with nipple adapters to hold the ampoules, a clamp, and an acrylic plate with a gasket.

ADSM 096. - 2 x 48 port manifolds with nipple adapters, manifolds connecting pipe, 3 clamps, and an acrylic plate with a gasket.

PRODUCT CODE	ITEM	CAPACITY
ADSM 024	24 - Port Secondary Drying Manifold	24 ampoules
ADSM 048	48 - Port Secondary Drying Manifold	48 ampoules
ADSM 096	96 - Port Secondary Drying Manifold	96 ampoules



LYODEL - FOOD LYOPHILISERS (SINGLE CHAMBER)

Lyodel Food Freeze Dryers (SINGLE CHAMBER) are the perfect option for freeze-drying fruits, vegetables, meats, dairy products, flowers, herbs and last but not least for small-scale lab applications.



TECHNICAL SPECIFICATION OF FOOD FREEZE DRYERS

PRODUCT CODE / MODEL NO	LFFD - 2	LFFD - 5	LFFD - 10
TYPE	FLOOR STANDING		
HEATING METHOD	THERMOSTATIC SHELF HEATING		
TRAY AREA (sqm)	0.3	0.7	1
TRAY SIZE (W X D X H) mm	145 X 275 X 20	230 x 500x20	350 X 500 X 20
NO OF TRAYS	4	5	6
SPACE BET TRAYS (mm)	60	50	50
LOADING CAPACITY (BULK) @ 10mm	1 - 2	3 - 5	6 - 10
ICE CAPACITY (L)	3	5	10
COMPRESSOR	1		
ULT. LOW TEMP (C)	-40°C (RT<250C)		
VACUUM PUMP TYPE	DIRECT DRIVE, DOUBLE STAGE, OIL SEALED		
VACUUM PUMP (LPM)	100	150	200
VACCUM DEGREE (mbar)	1000 - 0.200		
REFRIGERANT	CFC & HCFC FREE		
POWER CONSUMPTION (kWh)	1.25	2.5	3
DIMENSION (W X D X H) mm	510 X 540 X 900	560 X 720 X 920	800 X 750 X 1300
POWER SUPPLY	230V, 50Hz, 1P		

LYODEL - FOOD LYOPHILISERS (DOUBLE CHAMBER)

Lyodel Food Freeze Dryers (DOUBLE CHAMBER) are the perfect option for freeze-drying fruits, vegetables, meats, dairy products, flowers, herbs and last but not least for small-scale lab applications.



TECHNICAL SPECIFICATION OF FOOD FREEZE DRYERS

PRODUCT CODE / MODEL NO	LFFD - 20	LFFD - 30	LFFD - 40
TYPE	FLOOR STANDING		
HEATING METHOD	THERMOSTATIC SHELF HEATING		
TRAY AREA (sqm)	2.1	3.2	4.3
TRAY SIZE (W X D X H) mm	600 X 300 X 20	600 X 300 X 20	600 X 400 X 20
NO OF TRAYS	12	18	18
SPACE BET TRAYS (mm)	90	75	75
LOADING CAPACITY (BULK) @15mm	10 - 20	20 - 30	30 - 40
ICE CAPACITY (L)	20	30	40
COMPRESSOR	2		
ULT. LOW TEMP (C)	-40°C (RT<25C)		
VACUUM PUMP TYPE	DIRECT DRIVE, OIL SEALED		
VACUUM PUMP (LPM)	600	600	750
VACCUM DEGREE (mbar)	1000 - 0.200		
REFRIGERANT	CFC & HCFC FREE		
POWER CONSUMPTION (kWh)	6	15	20
DIMENSION (W X D X H) mm	2800 X 1500 X 2500		
POWER SUPPLY	440V, 50Hz, 3P		

LYODEL - TRAY LYOPHILISERS

Lyodel Tray Freeze Dryers (LTFD) are time crunchers and economical, providing basic freeze-drying without inbuilt freezing. LTFDs are equipped with heaters for faster completion of the process. Comes in both PID and HMI touch controls.

FEATURES

- 2 Cascade refrigeration system
- Automatic defrost with hot gas
- Removable trays
- MOC - SS 304
- Chamber vacuum Control is provided
- Transparent door for easy visibility of the product chamber.



TECHNICAL SPECIFICATION OF BASIC TRAY FREEZE DRYERS

PRODUCT CODE / MODEL NO	LTFD - 5	LTFD - 10	LTFD - 20
TYPE	FLOOR STANDING		
HEATING METHOD	THERMOSTATIC PAD HEATING		
NO OF RACKS	4	5	9
TRAY AREA (sqm)	0.5	1	2
TRAY SIZE (W X D X H) mm	390 X 350 X 25	390 X 490 X 20	390 X 570 X 20
NO OF TRAYS	4	5	9
TRAY TEMPERATURE	Ambient to +70C		
SPACE BET TRAYS (mm)	30		
MAX LOADING CAPACITY (kg)	5	10	20
ICE CAPACITY (kg)	5	10	20
COMPRESSOR	1 or 2		
CONDENSER TEMP +/-5C@20C amb	-50°C/ -80°C (RT<25C)		
VACUUM PUMP TYPE	DOUBLE STAGE, OIL SEALED		
VACUUM PUMP (LPM)	250	350	500
VACCUM DEGREE (mbar)	1 X 10 ⁻³		
HEATER TEMPERATURE SENSORS	1		
REFRIGERANT	CFC & HCFC FREE		
VACUUM GAUGE	DIGITAL PIRANI GAUGE		
TEMP INDICATORS	PID CONTROLLERS / HMI		
DEFROST	MANUAL / HOT GAS		
POWER SUPPLY	220V, 50Hz, 1P	440V, 50Hz, 3P	

LYODEL - SHELF FREEZE DRYERS

Lyodel Shelf Freeze Dryers (LSFD) are specially designed for flexible and reproducible production cycles with an inbuilt freezing technology. In LSFD, products can be lyophilised in flasks(optional), vials, and bulk with a heating facility.

With separate drying and condenser chambers, Shelf Freeze Dryers work with condenser temperatures of -50C / -80C. It comes with 1 / 2 / 3 shelves that are cooled through direct expansion of the refrigeration circuit and heated by an electrical resistance at low voltage. The refrigeration system consists of a single refrigerating unit using a hermetic compressor and non-CFC refrigerant, operating both on the shelf and condenser thereby guaranteeing system performance flexibility and safety. Comes in both PID and HMI touch controls.

FEATURES

- The distinct Stainless steel chamber and condenser chamber design provide hassle-free operation of the machine.
- INBUILT FREEZING
- Automatic defrost system
- High system performance flexibility and safety
- A wide range of accessories can be used
- Microprocessor fully automatic control system for precise controls
- Provision for Vacuum sealing
- Different container sizes can be sealed under a vacuum



TECHNICAL SPECIFICATION OF SHELF FREEZE DRYERS

TECHNICAL DATA	LYODEL SINGLE SHELF		LYODEL DOUBLE SHELF		LYODEL TRIPLE SHELF	
PRODUCT CODE / MODEL NO	LSFD 1S	LSFD 1SU	LSFD 2S	LSFD 2SU	LSFD 3S	LSFD 3SU
CONDENSER TEMP +/-5C@20C amb	-50 C	-80 C	-50 C	-80 C	-50 C	-80 C
HEATING METHOD	THERMOSTATIC PAD HEATERS / THERMAL OIL					
COND ICE CAPACITY (kg/24hr)	5					
TOTAL SHELVES (no)	1+1		2+1		3+1	
USABLE SHELVES (no)	1		2		3	
SHELF TEMP CONTROL (C+/-5C)	-45C +50C	-60C +60C	-45C +50C	-60C +60C	-45C +50C	-60C +60C
SHELF SIZE - W X D (mm)	275 X 350					
SHELF AREA (m2)	0.10		0.20		0.3	
DISTANCE BETWEEN SHELVES (mm)	135		80		65	
BULK CAPACITY - @ 10mm DEPTH	1 L (max)		2 L (max)		3 L (max)	
BULK CAPACITY - @ 20mm DEPTH	2 L (max)		4 L (max)		6 L (max)	
VIAL STOPPERING MECHANISM	MANUAL OR AUTOMATIC (OPTIONAL)					
MAX VIAL CAPACITY	2ml-638; 5ml-285; 10ml-21		2ml-1584; 5ml-864; 10ml-468		2ml-1914; 5ml-855; 10ml-63	
MAX FLASK CAPACITY	4 or 6 PORT SS MANIFOLD (OPTIONAL)					
REFRIGERANT	CFC & HCFC FREE					
VAC & TEMP INDICATION	MICROPROCESSOR CONTROL SYSTEM					
VACUUM RANGE (mbar)	1000 - 0.100					
DIMENSION - W x D x H (mm)	1200 X 750 X 1250/1600					
WEIGHT (kg)	300 - 400					
INPUT POWER SUPPLY	230V / 50Hz, 1 phase or 440VAC/50Hz, 3 phase, (3KW - 5KW max)					

LYODEL - ULTRA SHELF FREEZE DRYERS

Lyodel Ultra Shelf Freeze Dryers (USFD) are compact benchtop freeze dryers that come with a single-shelf internal cooling and heating technology for flexible and reproducible production cycles with an internal SS Condenser coil. With UNI drying and condenser chamber, Shelf Freeze Dryers work with condenser temperatures of -50°C / -85°C . The USFD comes in 4 variants for multiple applications.



FEATURES

- The distinct Stainless steel chamber and condenser chamber design provide hassle-free operation of the machine.
- INBUILT FREEZING
- Internal SS Condenser Coil
- Automatic defrost system
- High system performance flexibility and safety
- A wide range of accessories can be used (Bulk, Flask, Vials)
- Microprocessor fully automatic control system for precise controls
- Provision for Vacuum sealing
- Different vial sizes can be sealed under a vacuum

TECHNICAL SPECIFICATION OF ULTRA SHELF FREEZE DRYERS

TECHNICAL DATA	BENCHTOP - LYODEL SINGLE SHELF FREEZE DRYER			
PRODUCT CODE / MODEL NO	USFD	USFD - F	USFD - S	USFD - FS
CONDENSER TEMP $\pm 5^{\circ}\text{C}@20^{\circ}\text{C amb}$		-85		
CONDENSER VOLUME (L)		11		
COND ICE CAPACITY (kg/24hr)		3		
TOTAL SHELVES (no)		1+1		
USABLE SHELVES (no)		1		
SHELF TEMP CONTROL ($^{\circ}\text{C} \pm 5^{\circ}\text{C}$)		-60 to +70		
SHELF SIZE - W X D (mm)	270 X 400 X 150		270 X 400 X 120	
SHELF AREA (m ²)		0.10		
VACUUM PUMP		100 LPM		
BULK CAPACITY - @ 10mm DEPTH		3 L		
TEMPERATURE UNIFORMITY ($^{\circ}\text{C}$)		± 1		
VIAL STOPPERING MECHANISM		AUTOMATIC		
MAX VIAL CAPACITY	NA	NA	2ml-726; 5ml-400; 10ml-216	
MAX FLASK CAPACITY	NA	4	NA	4
REFRIGERANT		CFC & HCFC FREE		
VAC & TEMP INDICATION		MICROPROCESSOR TOUCH SCREEN		
VACUUM DEGREE (mbar)		0.01		
DIMENSION - W x D x H (mm)	820 x 695 x 600/995		820 x 695 x 875/1075	
WEIGHT (kg)	157	165	165	170
INPUT POWER SUPPLY		220V, 50Hz, 2.3KW		

LYODEL - PILOT LYOPHILISERS (- 50C)

Lyodel Pilot Freeze Dryers are high-capacity Shelf Freeze dryers with 4 - 6 shelves. These inbuilt freezing dryers are ideal for freeze-drying sensitive materials in vials or bulk. The machine is built with advanced technology electrical and electronics that give hassle-free operation for the user. The unit has an integral refrigeration system allowing precise temperature control.

SALIENT FEATURES

- Temperature uniformity on all shelves throughout the process
- INBUILT FREEZING
- Product safety assured with accurate temperature control to $\pm 1^{\circ}\text{C}$
- Silicone cooling fluid
- VIAL STOPPERING (OPTIONAL) under vacuum/inert gas
- Complete with full instrumentation for process optimisation and pressure rise testing
- Microprocessor fully automatic control system



TECHNICAL SPECIFICATION OF LYODEL PILOT FREEZE DRYERS (-50C)

Model	LSFD 5S	LSFD 5ST	LSFD 20S	LSFD 20ST	LSFD 40S	LSFD 40ST
Shelf temp (C)	-40 to 50		-40 to 50		-40 to 50	-40 to 50
Final condenser temp (C)	-50		-50		-50	-50
Vacuum Degree (mbar)	<10		<10		<10	<10
Vacuum Pump * (LPM)	250		500		1000	
Freeze drying area (m2)	0.5		1		2	
Condenser capacity (kg/24h)	8		15		30	
Shelf (L x W x H)mm	250 X 400 X 20		550 X 480 X 20		560x600x20	
Qty of shelf	5+1		4+ 1		6+1	
Distance between shelves(mm)	65		90			
Bulk Loading capacity @ 10mm (L)	5		10		20	
Bulk Loading capacity @ 20mm (L)	10		20		40	
Condensor Cooling mode	Air Cooling			Water cooling		
Electric heater defrost	Y		Y		Y	Y
Stoppering Mechanism	NA	YES	NA	YES	NA	YES
Penicillin bottle (Dia 22)		1040		2016		4100
Penicillin bottle (Dia 16)		1944		3944		7800
Penicillin bottle (Dia 12)		3456		7020		14400
Power supply (V/Hz)	440V/50Hz, 3P					
Exterior Dimension (WxDxH)mm	1350x850x1650/2000		1500x1000x1600/1950		2100x1200x1800/2100	

* Imported Vacuum Pump - Optional

LYODEL - ULTRA PILOT LYOPHILISERS (-80C)

Lyodel Ultra Pilot Freeze Dryers are -80C dryers used in places where the samples are sensitive and the freezing point is above -40C. The machine is built with advanced technology electrical and electronics that gives hassle-free operation for the user. The unit has an integral refrigeration system allowing precise temperature control.

SALIENT FEATURES

- Temperature uniformity on all shelves throughout the process
- INBUILT FREEZING
- Product safety assured with accurate temperature control to $\pm 1^{\circ}\text{C}$
- Silicone cooling fluid
- VIAL STOPPERING (OPTIONAL) under vacuum/ inert gas
- Complete with full instrumentation for process optimisation and pressure rise testing
- Microprocessor fully automatic control system



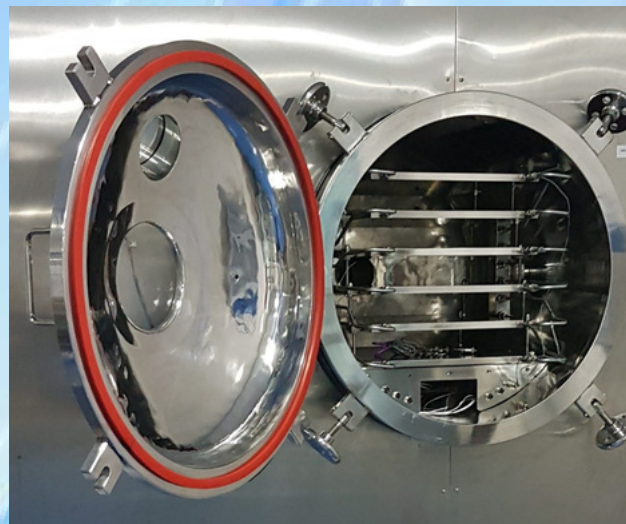
TECHNICAL SPECIFICATION OF LYODEL ULTRA PILOT FREEZE DRYERS (-80C)

Model	LSFD 5SU	LSFD 5SUT	LSFD 10SU	LSFD 10SUT	LSFD 20SU	LSFD 20SUT
Shelf temp (C)	-5o to 7o		-5o to 7o		-5o to 7o	-5o to 7o
Final condenser temp (C)	-80		-80		-80	-80
Vacuum Degree (mbar)	<10		<10		<10	<10
Vacuum Pump * (LPM)	250		500		1000	
Freeze drying area (m2)	0.5		1		2	
Condenser capacity (Kg/24h)	10		20		40	
Shelf (L x W x H)mm	400 X 250 X 16		550 X 480 X 18		600 X 560 X 20	
Qty of shelf	5+1		4+ 1		6+1	
Distance between shelves(mm)	65				90	
Bulk Loading capacity @ 10mm (L)	5		10		20	
Bulk Loading capacity @ 20mm (L)	10		20		40	
Condensor Cooling mode	Air Cooling				Water cooling	
Hot Gas defrost	Y		Y		Y	Y
Stoppering Mechanism	NA	YES	NA	YES	NA	YES
Penicillin bottle (Dia 22)		1040		2016		4100
Penicillin bottle (Dia 16)		1944		3944		7800
Penicillin bottle (Dia 12)		3456		7020		14400
Power supply (V/Hz)	440V/50Hz, 3P					
Exterior Dimension (WxDxH)mm	1350x850x1650/2000		1500x1000x1600/1950		2100x1200x1800/2100	

* Imported Vacuum Pump - Optional

LYODEL - 1sqm LYOPHILIZERS

These are custom-built freeze dryers manufactured as per user requirements. The equipment is equipped with highly advanced electrical and electronics controls, fully automatic with SIP / CIP provision and placed in Clean room.



DIRECT DRIVE HIGH VACUUM PUMPS

DELVAC 'S Direct Driven High Vacuum Pumps are designed to meet the requirement of high vacuum applications of Freeze Drying. The pump comprises of double-stage offset rotary vane arrangement with a special inbuilt oil pump to take care of oil circulation and to assure continuous silent operation under high vacuum. Operation of the pump with the inbuilt gas ballast valve in open condition warms up the pump thereby preventing vapour condensation inside the pump. This increases the water handling capacity and thereby makes them ideal for use in Freeze Drying operations. The advanced lubrication system with a special in-built oil pump enables the vacuum pumps to work with wide oil level tolerance, reducing the need for frequent topping up. These pumps are modular in construction which makes the dismantling and assembling hassle free.

SPECIAL FEATURES

- Built-in anti-suck-back mechanism
- Built-in gas ballast
- Silent operation
- High vapour handling capacity
- Low oil consumption
- Compact and sturdy

TECHNICAL SPECIFICATION OF DELVAC'S VACUUM PUMP			
TECHNICAL DATA	DD 8	DD 12	DD 18
PRODUCT CODE	VP 001	VP 002	VP 003
DISPLACEMENT (L/min)	100	200	300
MAX. VACUUM (mbar)	0.03	0.03	0.05
MAX WATER VAPOUR INLET PRESSURE (mbar)	15	15	20
MAX WATER VAPOUR PUMPING RATE (kg/hr)	0.06	0.15	0.30
OIL CAPACITY (L)	0.5	0.5 - 0.75	0.75 - 1
RECOMMENDED OIL	MOLECULAR DISTILLED VACUUM OIL		
OIL MIST FILTER	EMF 10	EMF 10	EMF 20

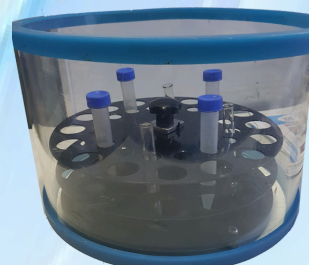
PRE - FREEZING GLYCOL BATH

Pre-freezing Glycol bath is meant to freeze materials in round bottom flasks before loading them on the freeze dryer. The pre-freezer bath can achieve a temperature range of -40°C. The rotation mechanism of the pre-freeze bath helps the sample in the flask to freeze to a thin layer on the inner walls of the flask which promotes faster drying and reduce time.

PRODUCT CODE	ITEM	RPM	POWER
FMFB 040	Pre - freezing Glycol bath	100	220V, 50Hz, 1ph

OTHER SUPPORTING ACCESSORIES FOR FREEZE DRYERS

PRODUCT CODE	ITEM	MOC
FREEZE DRYING - FLASK		
FMV 034	Spare ¾" quick seal valve with knob	RUBBER / PLASTIC
FMPC	Connector for quick seal valve and adaptor	PLASTIC
FMCCS FMCCM FMCCCL	Cone connectors for Flasks -small B19 Cone connectors for Flasks -medium B24 Cone connectors for Flasks -large B29	RUBBER
RBF 0050 RBF 0100 RBF 0250 RBF 0500 RBF 1000	50 ml Round bottom flasks with std. neck size B19 100 ml Round bottom flasks with std. neck size B19 250 ml Round bottom flasks with std. neck size B24 500 ml Round bottom flasks with std. neck size B29 1000 ml Round bottom flasks with std. neck size B29	BOROSILICATE GLASS
WMF 040 WMF 080 WMF 150 WMF 300	40 ml Wide mouth Flask w cap - working capacity 20ml 80 ml Wide mouth Flask w cap - working capacity 40ml 150 ml Wide mouth Flask w cap - working capacity 75ml 300 ml Wide mouth Flask w cap - working capacity 150ml	BORO SILICATE GLASS
FREEZE DRYING - AMPOULES		
ADST 100	Spark tester - Leak checking of ampoules	
ADST 101	Ampoule sealing torch - Sealing ampoules under a vacuum	
ADST 102	Adapter for 12 ampoules (pre-freeze separately) can be used along with flask manifold	
FDA 000 FDA 002 FDA 005 FDA 010	Freeze drying ampoules 0.5 ml (bulb bottom) - 1000 no Freeze drying ampoules 2 ml (bulb bottom) - 1000 no Freeze drying ampoules 5 ml (bulb bottom) - 1000 no Freeze drying ampoules 10 ml (bulb bottom) - 1000 no	GLASS
FREEZE DRYING - VIALS		
FDV 002 FDV 005	2 ml Tubular vials - 5000 no 5 ml Tubular vials - 5000 no	GLASS
VSA 002 VSA 005	Stoppers for 2 ml vials Stoppers for 5 ml vials	RUBBER
VCA 002 VCA 005	Caps for 2 ml vials Caps for 5 ml vials	ALUMINIUM
OTHER COMMON ITEMS		
MDVO	Vacuum Oil 5L - 50 L	
SGL 200 SGL 215 SGL 250 SGL 300 SGL 310	Sealing gasket (Ø200) Sealing gasket (Ø215) Sealing gasket (Ø250) Sealing gasket (Ø300) Sealing gasket (Ø310)	SILICONE
VSIV 01	Isolation valve to separate machine and vacuum pump (additional safety to prevent oil entering chamber)	



FREEZE DRYING APPLICATIONS

FOOD INDUSTRY - Instant foods, fruits, veggies, meat, fish etc

DAIRY INDUSTRY - Lactoferrin, ice creams, cheese etc

CHEMICAL INDUSTRY - Chemical synthesis

NUTRACEUTICALS - Increase the shelf life of products

BIOTECHNOLOGY - Bio separations, enzymes, blood plasma,

NANOTECHNOLOGY - Nanotube Purification

STATERS & CULTURES - Probiotics and buttermilk

NASA & MILITARY - Astronaut meals and Military rations

PHARMACEUTICALS - Preparation of vaccines and drugs

RESEARCH - Laboratory samples, microbial cultures etc

FLORAL. - Preserve Flowers with structure and quality

TAXIDERMY - Preserve all types of animals, insects,

DOCUMENT RECOVERY - Recover water-damaged books and documents

OUR CUSTOMERS



AND MORE...



DELVAC
PUMPS PVT. LTD.



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OTHER PRODUCTS WE OFFER: VACUUM FRYING MACHINES AND COMPLETE ACCESSORIES FOR MAKING FRUIT, VEGETABLE, NUTS, PULSES SNACKS